

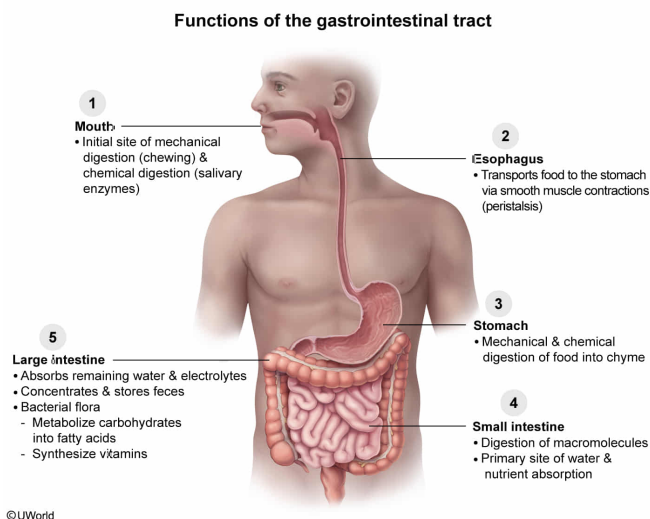
During a nutrition study, a participant consumes five times the physician-recommended amount of food but is unable to sufficiently absorb nutrients. Given this information, this participant would be LEAST likely to exhibit which of the following conditions?

- ☐ A. Loss in diversity of bacteria that produce fatty acids in the large intestine [8%]
- ☐ B. Decreased surface area in the small intestine [7%]
- ☐ C. Heightened production of serum antibodies against intestinal proteins [6%]
- ✓ ☒ D. Increased activity of nutrient transporter proteins in the small intestine [77%]

Correct

77% answered correctly

Explanation:



The gastrointestinal tract is composed of the esophagus, stomach, small intestine, and large intestine. After undergoing initial breakdown in the mouth, ingested food begins its passage through the gastrointestinal tract by traveling down the esophagus to the stomach where chyme, a semifluid mixture of partially digested food, is formed. This chyme then exits the stomach to begin transit through the intestines.

First, in the small intestine, intestinal and **pancreatic** enzymes break down **macromolecules**, which are then absorbed via diffusion through absorptive cell membranes or through intestinal transporters. Next, food enters the large intestine, the site where additional water absorption occurs and where microbial

communities (gut flora) synthesize vitamins and metabolize remaining carbohydrates.

In this scenario, the participant consumes a large meal but is unable to sufficiently absorb nutrients (ie, exhibits malabsorption). Nutrient absorption in the gastrointestinal tract is influenced by several factors, including:

- The presence and diversity of fatty acid–producing bacteria in the large intestine. These bacteria **metabolize** undigested carbohydrates from the small intestine into short-chain fatty acids, which are absorbed and used as additional energy sources. A *decrease* in these bacterial species would lead to *decreased* absorption of additional fatty acids as nutrients **(Choice A)**.
- The small intestine's **surface area**. Large folds in the intestine's **epithelial** lining slow the movement of chyme through the intestinal tract. In addition to these large folds, villi (fingerlike projections extending from the lining) and microvilli (smaller fingerlike extensions of individual absorptive cells) *maximize* the time and surface area available for nutrient absorption. Malabsorption is sometimes caused by structural damage (eg, villi destruction) that decreases small intestinal surface area, which thereby *impairs* nutrient absorption **(Choice B)**.
- The functions of intestinal proteins. Structural proteins, digestive enzymes, and nutrient transporters work synergistically to break down and absorb nutrients within the gastrointestinal tract. Because **antibodies** mark cellular materials for destruction by cells of the immune system, production of antibodies *against* intestinal proteins would likely cause the destruction of these proteins, damaging the intestinal tract and *decreasing* nutrient absorption **(Choice C)**.

Intestinal nutrient transporters facilitate nutrient absorption from the intestinal tract into the body. Therefore, **increased** nutrient **transporter activity would increase** (*not* decrease) **nutrient absorption**.

Educational objective:

Nutrient absorption in the gastrointestinal tract is affected by the diversity of intestinal bacteria (gut flora), the surface area of the small intestine, and the functions of intestinal proteins (digestive enzymes, nutrient transporters, structural proteins).